

App Development Project Report

Trekking Management App

1. Student Details

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About Me:- I am MD TOKIR RAZA, a Diploma-level student in the BS in Data Science program at IIIT Madras. I have a strong interest in web application development and cybersecurity. I really enjoy working on projects in these areas and continuously strive to improve my skills through hands-on learning.

2. Project Details

Project Title: Trekking Management App

Problem Statement:

Managing trekking activities manually is time-consuming and prone to errors such as overbooking, poor record management, and difficulty in tracking participants. This project provides a web-based Trekking Management System where admins can manage treks and staff, staff can update trek information, and trekkers can search, book, and track their trekking history efficiently.

Approach:

The project is developed using the Flask framework with SQLite as the database. It follows a role-based approach where Admin, Staff, and Trekker have separate dashboards and permissions to manage treks, bookings, users, and staff efficiently.

3. AI/LLM Declaration

I used **ChatGPT (GPT-5)** assist in understanding concepts, writing SQLAlchemy model definitions, generating documentation and mostly improving user interface of page(mostly css). The extent of AI/LLM usage is around **10–15%**, limited to **code suggestions and**

documentation formatting. All final implementation logic, debugging, and integration were done manually.

4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Core backend web framework
SQLAlchemy	Object Relational Mapper for SQLite database
Jinja2	Template engine for rendering dynamic HTML pages
HTML/CSS	Create the structure of web pages and Frontend styling and responsive design.
SQLite	Lightweight local database for storing user data
Flask-Login	User authentication and session management
Python	Used as the main programming language for backend development.
Visual Studio Code	Used as the code editor for developmen

5. Database Schema / ER Diagram

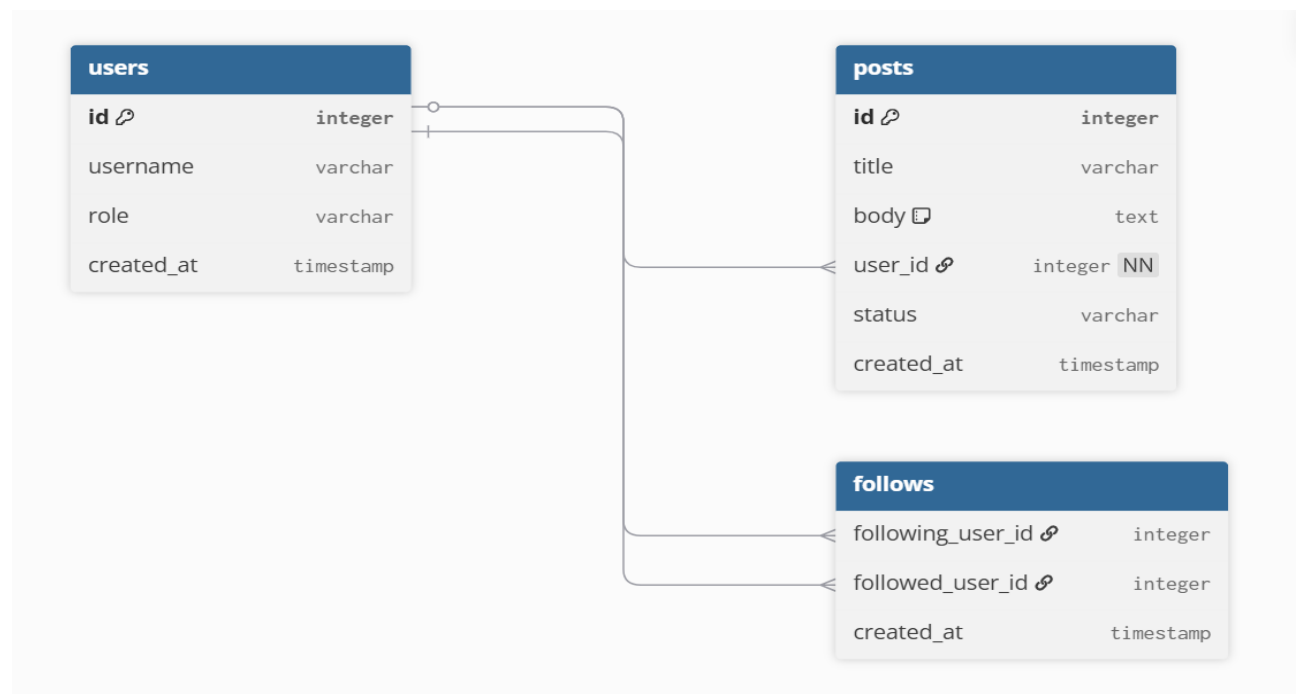
Tables:

1. **User** — Stores user information such as Admin, Staff, and Trekker details, login credentials, approval, and blacklist status.
2. **Trek** — Stores trek details including trek name, location, difficulty, duration, available slots, dates, status, and assigned staff.
3. **Booking** — Stores booking information such as booking ID, user ID, trek ID, booking date, trek name, user name, and booking status.

Relationships:

- One-to-Many → User → Booking
- One-to-Many → Trek → Booking
- One-to-One → Staff → Trek

(Insert ER diagram image here – can be created using dbdiagram.io or draw.io or any other website)



(example taken from dbdiagram.io)

6. API Resource Endpoints

Endpoint	Method	Description
/	GET	Display Home Page
/login , /register	GET, POST	User login . register a new user

<code>/cancel_booking/<int:booking_id></code>	POST	Cancel Booking , Assign Staff to Trek , Approve Staff, Blacklist User , Blacklist Staff
<code>/blacklist_staff/<int:u_id></code>		etc.
<code>/booking/<int:trek_id></code>	GET , POST	Book a Trek , Edit Trek Details and Delete Trek
<code>/edit_trek/<int:trek_id></code>		
<code>/delete_trek/<int:trek_id></code>		

7. Architecture and Features (optional)

Architecture Overview :- The project follows a three-tier architecture consisting of the .Presentation Layer (HTML, CSS, Jinja2), Application Layer (Flask), and Database Layer (SQLite with SQLAlchemy).

- **app.py** – main Flask application entry point
- **/instance** – database models using SQLAlchemy
- **/templates** – Jinja2 HTML templates
- **/static** – CSS, JS, and img-logo files

Implemented Features:

- Role-based login for Admin, Staff, and Trekker.
- Trek management and staff assignment.
- Trek search and online booking.
- Booking cancellation with automatic slot update.
- User, staff, and booking management.
- Approval and blacklist functionality for staff and users.

8. Video Presentation

Drive:

https://drive.google.com/file/d/17HHYDVUGGOEiLFjDU_CAE7riVZ2h00k1/view?usp=sharing

Thank You.